

# MAYOPO ADEOYE

## Full Stack Developer (Fintech & Enterprise Systems)

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## PROFESSIONAL SUMMARY

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Full Stack Developer with 5+ years of experience building and shipping production systems for live banking and fintech clients, including a foreign exchange trading platform used daily by Access Bank Africa traders. Skilled across React, Node.js/NestJS, Java/Spring Boot, and .NET, with hands-on experience in real-time systems (WebSocket/STOMP), secure payment integrations (Stripe, Paystack, Twilio), and enterprise authentication (Active Directory, JWT, RBAC). Scored 4.85/5.00 (Strong Hire) on an enterprise payments engineering assessment covering double-entry ledger design and transaction concurrency. Comfortable owning a product end-to-end or operating within a structured engineering team.

## TECHNICAL SKILLS

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- **Languages:** JavaScript, TypeScript, Python, C#, Java, Solidity
- **Frontend:** React, Vue, Quasar (Capacitor), Redux Toolkit, Material UI, Tailwind CSS, SASS
- **Backend:** Node.js, Express.js, NestJS, Java Spring Boot, .NET, REST, GraphQL
- **Databases:** PostgreSQL, MongoDB, MySQL, OracleDB, Redis
- **DevOps & Tools:** Git, GitHub, GitLab, Docker, Render, Postman, Swagger/OpenAPI, Prometheus
- **Integrations & Security:** Stripe, Paystack, Twilio, Socket.IO, Cloudinary, Active Directory SSO, JWT, OAuth, OWASP
- **System Design & Security:** Double-Entry Ledger Design, Concurrency Control (Row/Advisory Locking), Distributed Tracing, Event-Driven Architecture, Webhook Security (HMAC-SHA256).

## PROFESSIONAL EXPERIENCE

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### Software Engineer (Node.js and React) - Aristack Solutions (RetailFX - Access Bank Africa)

Jul 2023 – Jul 2026

- Core engineer on a live FX trading platform used daily by Access Bank Africa traders, handling real-money movements and regulatory compliance requirements.
- Architected role-based interfaces (Admin, Maker, Checker) with Active Directory SSO, and Engineered a concurrent bulk trade upload pipeline utilizing batched transactional OracleDB commits, reducing system lock contention and slashing trade booking latency by **60%**
- Implemented Hold & Release controls via Amount Block Numbers and a CRON-based status sync to prevent unauthorized fund movement and keep trade statuses current in real time.
- Developed a system-wide audit trail capturing user actions, IP, and MAC addresses, used directly for fraud investigation and regulatory reporting; built real-time quote updates via WebSocket and trade reversal workflows with status-based validation.

*Tech:* React, TypeScript, Node.js, Express.js, OracleDB, Active Directory, WebSocket, Calypso, Flexcube, Java, Git

### Software Engineer (Java Spring Boot & React) - Collaboration - Treasury Management System

2025

- Contributed to an enterprise Treasury Management System supporting core trading, asset management, and internal financial controls, with business workflow automation via Flowable BPMN for multi-stage approvals.
- Built role-based access control and multi-level authorization using Spring Security, JWT, and RBAC policies.
- Developed real-time notification and approval workflows using WebSockets (STOMP/SockJS), enabling task-claiming across treasury roles with email alerts via Spring Mail.
- Built a secure document upload pipeline with content extraction into Oracle Database, and implemented monitoring/metrics with Spring Boot Actuator and Prometheus.

*Tech:* Java 21, Spring Boot 3, Spring Security, Flowable BPMN, WebSockets (STOMP/SockJS), Oracle Database, PostgreSQL, React, Redux Toolkit, GoJS, Formik/Yup

### Lead Full Stack Developer (MERN Stack) - Freelance / Contract - Scholist (University Scholarship Discovery Platform)

Mar 2023 – Present

- Led end-to-end development of a scholarship discovery platform serving university students, improving discovery speed by 40%.
- Built a Cheerio-based scraper that auto-indexes new scholarships in real time, expanding the live database by **50%**.
- Built an advanced filter engine (deadline, GPA, major, citizenship) that cut average search time by **35%**.
- Integrated Stripe payments and Socket.IO coaching messages, boosting student engagement by **30%**.

*Tech: React, Node.js, MongoDB, Redux, Socket.IO, Stripe, Cheerio, JavaScript*

## SELECTED PROJECTS

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### Payments & Ledger Engine - *Technical Assessment*

2026

- Built a full-stack payment processing system (Node.js/Express, raw pg driver, PostgreSQL, React) implementing a double-entry ledger with BIGINT minor-unit accounting, per-currency transit accounts, and point-in-time running-balance snapshots.
- Designed a triple-layer concurrency model (Postgres advisory locks, SELECT FOR UPDATE, SERIALIZABLE isolation) enforcing deterministic lock ordering to guarantee strictly serializable, deadlock-free execution under high-concurrency stress tests. .
- Implemented a webhook processing pipeline with raw-body HMAC-SHA256 verification, timing-safe signature comparison, and a validated transaction-state-machine for settlement and reversal.
- Built an FX quoting service with configurable spread, rate locking, and TTL-based expiry, plus an idempotency layer for safe payment retries; scored 4.85/5.00 with a “Strong Hire” recommendation, with reviewer feedback praising the concurrency design as “exceptionally rigorous” and the ledger schema as a “production-grade ledger model.”

*Tech: Node.js, Express.js, PostgreSQL, React, HMAC-SHA256, Jest*

### Ibloom - *Chair Rental & Booking Platform*

2025

- Built and deployed an end-to-end booking and logistics platform for a rental business, live with paying customers.
- Delivered order management, admin operations, real-time chat (Socket.IO), and automated transactional emails.

*Tech: Node.js, Express.js, Redux Toolkit, Tailwind CSS, Cloudinary, Socket.IO, Nodemailer*

### Chronos - *Multi-Product Order Management System*

2025

- Built a NestJS + PostgreSQL REST API with a Quasar/Capacitor frontend, supporting both web and native mobile deployment.
- Integrated Twilio SMS and Nodemailer for order notifications; delivered a full client handover including GitLab repos, Postman collection, Render deployment, and a technical decision memo.

*Tech: NestJS, TypeScript, PostgreSQL, Quasar, Capacitor, Twilio, GitLab, Render*

### Fintech Payment Gateway - *Kuda-style Platform*

2024

- Built a full .NET/C# payment platform covering registration, account balances, deposits, transfers, and real-time processing.
- Secured the platform with Paystack integration, Twilio OTP verification, and Redis-based session management.

*Tech: C#, .NET, Entity Framework, SQL, MongoDB, Redis, Paystack, Twilio, Swagger*

## EDUCATION & CERTIFICATIONS

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B.Sc. Mathematics - University of Lagos

Smart Contracts Certificate - State University of New York (SUNY) via Coursera, 2022

Languages: English (Fluent), Yoruba (Native)

## AWARDS & HONOURS

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- Strong Hire (4.85/5.00) - Enterprise Payments & Ledger Engineering Technical Assessment (2026)
- Top 7% (Silver Award) - International Youth Math Challenge (IYMC), 65+ countries (2020)
- Overall Best Student - NIIT e-Learning Scholarship (2018)
- 2nd Place - Nigerian Mathematics Olympiad (2016)
- 3rd Place - American Mathematics Olympiad (2017)
- Best Student - Covenant University Initiative Competition, Ogun State (2017)